

Immersive Virtual Reality Application with 3D Audio-Visual Scene Reconstruction from Single 360° RGB-D

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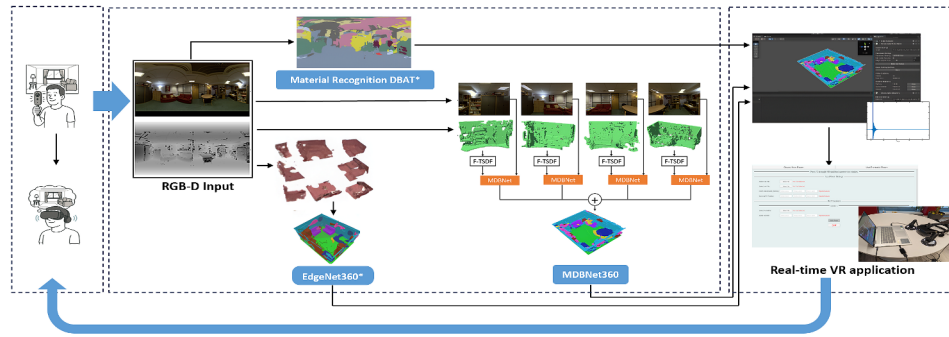


Figure 1: Illustration of VR system with single 360° RGB-D input. The system incorporate DBAT [5], EdgNet360 [6], and MDBNet360 pipelines and real-time VR application.

ABSTRACT

We propose a virtual reality (VR) system that enables users to experience an immersive 3D environment with full 360° surroundings and spatial sound in real time from a single 2D 360 RGB-D input. The system includes a 3D semantic scene completion and material recognition pipelines, and a Unity3D-based VR application for audio rendering with material geometry within the scene. Demo access is available at: <https://github.com/MonaIA1/Repo360>

Index Terms: Virtual Reality, Human-Centred Interactions, Immersive Environments.

1 INTRODUCTION

Today, perspective and 360° cameras are widely used to capture large amounts of RGB and depth data. This research focuses on designing a virtual reality (VR) application that transforms 2D images into comprehensive 3D models enriched with semantic and acoustics for immersive virtual environments. Artificial intelligence (AI) enables the inference of both structural and semantic labels of the 3D world, as well as material properties, from a single 2D RGB-D input. Within VR, users can interact with a simulated 3D environment in real time, experiencing a compelling sense of immersion in synthetic environments [8]. In this work, we propose a horizontally integrated AI-VR system designed to foster intuitive and human-centered digital interactions.

Prior work has explored audio-visual scene generation using different methods. [7] applied SegNet [3] to extract 2D semantics and projected them into 3D using depth information, and reconstructing geometry from clustered point clouds. Later, Kim et al. [6] and Alawadh et al. [2] used EdgeNet360 [6, 4] to perform semantic scene completion (SSC) in 3D space. Our work presents an audio-visual application that combines the reconstruction of 3D semantic scene and materials estimation from a single 360° RGB-D input,

allowing virtual environments to reflect both the visual layout and acoustic behavior of real-world scenes including the improvement on SSC part. We extend the SSC model in [1] with depth only input and propose MDBNet with RGB-D input then we extend the MDBNet inference on 360° RGB-D by proposing MDBNet360. We contribute by proposing VR application include different SSC models such as EdgeNet360 and MDBNet360 to construct scenes with full surrounding and plausible audio. Also, we provide subjective evaluation to evaluate the user experience within VR space. This work demonstrates the feasibility of 3D semantic reconstruction and completion for full panorama scenes, addressing the challenge of missing parallax in single center 360° captures.

2 PROPOSED SYSTEM

The proposed system aims to construct VR space from a single 360° RGB-D input. The system integrates different pipelines and provides a solution for dynamically creating immersive VR environment. We used CVSSP dataset¹ which consists of five indoor scenes with 360° RGB-D and ground-truth acoustic parameter measurements [7, 6]. The components of the system are the following:

2.1 Semantic Scene Completion (SSC)

We extend the SSC model with depth only input in [1] with a dual-head and combined loss function to train the model simultaneously with RGB and depth data of perspective views and propose MDBNet architecture. Then we extend the inference of SSC 3D model from partial views by MDBNet to full 360° coverage by proposing MDBNet360, enabling the generation of 3D annotated models with 360° view from a single full panorama RGB-D image. We employed spherical-to-cubic projection and 3D rotation for depth point clouds to ensure geometric consistency of depth data with cubic RGB views. The output of this pipeline is a 3D model for an indoor scene with full surroundings including semantic annotation of both visible and occluded regions, as shown in Figure 1. In our proposed application we also combine the construction by EdgNet360 where the 3D models are reconstructed by projecting all depth points into a 3D space. In order to cover the whole 360 surroundings, the 3D

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¹<http://3dkim.com/research/VR/index.html> (accessed in 2025)

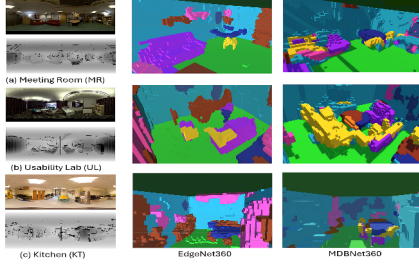


Figure 2: 3D SSC reconstructions by EdgeNet360 and MDBNet360 on CVSSP scenes.

coordinate is partitioned into eight overlapped view parts from the scene center. EdgeNet [4] is applied to individual areas for SSC inference and the output merged into one complete scene. Figure 2 shows 3D scenes constructed using EdgeNet360 and MDBNet360 SSC models. The final inferred 3D SSC model shows the scene reconstruction with semantic labels, and these labels are replaced by material labels inferred by the materials recognition module.

2.2 Material Recognition

We employ DBAT model by Heng et al.[5] to infer the materials categories and avoid the strong assumption of materials within the scenes. The authors propose dynamic backward attention transformer to aggregate cross-resolution features. The material labels were assigned to the 3D semantic labels within Unity platform ².

2.3 Audio-Visual Rendering in VR Space

The reconstructed 3D semantic scene is imported into Unity, where the Steam Audio plug-in ³ is used for room acoustics simulation and sound rendering. In each scene a virtual sound source and listener are placed within the scene, with sound rendering including the use of head-related transfer functions (HRTFs). Sound is generated in real time, adapting dynamically to the user's position and movement relative to the source, thereby enhancing immersion.

3 VR APPLICATION

The application designed based on the proposed system described above. The application includes the CVSSP RGB-D inputs. Users can select SSC reconstructed 360° scenes generated by either EdgeNet360 or by our proposed MDBNet360. The aim of the proposed application is to simplify room setup within Unity. A graphical user interface (GUI) was developed for users with limited familiarity with Unity's development tools.

For VR demonstration, an HP Reverb G2 headset and its controllers are used to navigate and interact with the virtual environment. The VR application uses the XR Interaction Toolkit 3.0.3 to manage controllers or hands and supports both direct (touch-based) and ray-based (distance) interactions with virtual objects. Also, it integrates smooth locomotion and teleportation for user movement. Smooth locomotion, is controlled by the left controller's analogue stick for fluid movement. It's also includes a dynamic FOV vignette that activates and adjusts dynamically based on the movement speed to reduce motion sickness. While teleportation, can be accessed through the right controller. It allows users to point to a location on the floor and instantly move there. User interaction is further supported by the XR Interaction Toolkit's affordance system, which provides visual cues for interactive objects. As an example, users can grab and manipulate the audio-source sphere in the scene.

²<https://unity.com/>(accessed in 2025)

³<https://valvesoftware.github.io/steam-audio/>(accessed in 2025)

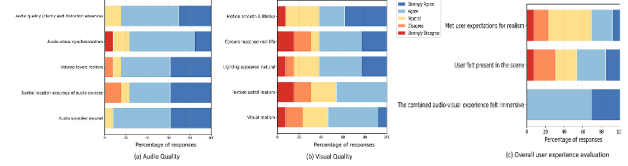


Figure 3: Participant ratings for the quality of audio (a) and visual (b) aspects, and overall experience (c).

3.1 Subjective evaluation

A user study was conducted to assess the overall effectiveness in delivering an immersive audio–visual experience. The participants were briefed, screened for VR suitability, and provided consent. All participants explored the same VR 3D scenes constructed by EdgeNet360 and then completed a five-point Likert-scale questionnaire assessing audio realism, visual fidelity, and overall experience, with optional open-ended feedback. Thirteen responses were completed the VR evaluation. Overall, participants rated the system as highly immersive as shown in Figure 3, with strong agreement on the clarity, spatial accuracy, and naturalness of the audio rendering. Visual feedback was positive but more mixed, with limitations in texture details and the realism of the represented scene. This is largely due to limitation in depth resolution resolution and mesh reconstruction. Nevertheless, participants felt that the integrated audio and visuals delivered a strong immersive effect. Qualitative comments further highlighted accurate spatial audio behaviour (e.g., distance attenuation and reverberation changes) as a key strength. Visual suggestions focused on improving geometry, texture resolution, and lighting. These findings confirm that the system delivers strong audio-driven immersion, while future work should enhance visual components to achieve better realism and presence.

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